

EM 601: Materials for Energy Applications

Course Code: EM 601

Course Name: Materials for Energy Applications

L-T-P-C: 3-1-0-4

Intended for: M.Tech Energy Materials

Distribution: Core course (Semester I)

Approval: 8th Senate (as EM 504); one-time approval as EM 601 (4 credits) in the 14th Senate (recommended by the 18th BoA)

Course Modules

- **Module 1:** Creep resistance materials: principles of creep deformation, mathematical modeling and design application of creep resistance materials in power plants. Wear resistance materials: types of wear, mechanism of erosion and cavitation wear, application of wear resistance materials in power plants, Fatigue types, types and protection. Life prediction of turbine materials. (10)
- **Module 2:** Conductors, transformer materials, Semi-conductors and devices, Fundamentals and energy applications for Magnetic and Dielectric materials. (16)
- **Module 3:** Energy efficient structure/building materials (8)

Text Books

1. Hans Berns, Werner Theisen, Gillian Scheibelein, *Ferrous Materials*, ISBN-10: 3540718478, 2008 (Springer).
2. A. J. Moulson and J. M. Herbert, *Electroceramics: Materials, Properties and Applications*, Wiley; 2nd edition, 2003.
3. M. Hall, *Materials for Energy Efficiency and Thermal Comfort in Buildings*, Elsevier Press, 2010.

Reference articles/books

1. Oliver Gutfleisch, Matthew A. Willard, Ekkes Brück, Christina H. Chen, S. G. Sankar, and J. Ping Liu, Magnetic Materials and Devices for the 21st Century: Stronger, Lighter, and More Energy Efficient, *Adv. Mater.* 23, 821-842 (2011)
2. L.H. Lewis and F.J. Villacorta, Perspectives on Permanent Magnetic Materials for Energy Conversion and Power Generation, *Metallurgical and Mater. Trans. A*, 44A, S1-S20 (2013)
3. S.L. Swartz, Topics in Electronic Ceramics, *IEEE Trans. Elect. Insulation*, 25 [5] 935 (1990)